



Wei-Jin Huang

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👤 Research Profile

- **Research Focus:** Ph.D. student at Sun Yat-sen University, advised by Prof. Wei-Shi Zheng, working on **embodied algorithms for humanoid robots and video action understanding**.
- **Internship Interests:** Combining VLA/WAM models, agentic planning, and continual learning with egocentric human data and robot experience for compositional generalization, rapid adaptation, and failure recovery.

🎓 Education

Sun Yat-sen University (Ph.D.) School of Computer Science and Engineering Sep. 2024 – Present
Computer Science and Research: Embodied Intelligence and Computer Vision Advisor: Prof. Wei-Shi Zheng
Technology (Changjiang Scholar)

- **Honors:** Sun Yat-sen University President's Scholarship (top 5%)
- **Publications:** Multiple first/co-first-author papers at CVPR, ECCV, and ACM MM.

South China University of Technology (B.E.) School of Computer Science and Engineering Sep. 2020 – Jul. 2024
Network Engineering Major Rank: 3/66

- **Honors:** National Scholarship (1/66), First-Class Tencent Scholarship (1/66)
- **Competitions:** Kaggle H&M Personalized Fashion Recommendations, Silver (85/2952, 2022); Kaggle G2Net Gravitational Wave Detection, Silver (57/1219, 2021)

📖 Selected Publications

Beyond Mimicry: Learning Whole-Body Human-Humanoid Interaction from Human-Human Demonstrations
IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR 2026)

Wei-Jin Huang, Yue-Yi Zhang, Yi-Lin Wei, Zhi-Wei Xia, Juantao Tan, Yuan-Ming Li, Zhilin Zhao, Wei-Shi Zheng

- **Overview:** Despite abundant human-human data, high-quality human-humanoid demonstrations are scarce: retargeting breaks key contacts, while standard imitation favors trajectory reproduction over interaction reasoning. PAIR generates executable supervision via contact-preserving, physics-aware retargeting. D-STAR separates *when/where*: Phase Attention models stages, Multi-Scale Spatial locates targets, and a diffusion head fuses both into synchronized interactions.

ChoreoPlan: Hybrid Phrase Planning and Execution-Grounded Selection for Music-to-Humanoid Dance
ACM International Conference on Multimedia (ACM MM 2026)

Wei-Jin Huang, Jianhong Fan, Hao Huang, Zhi-Wei Xia, Jun-Yi Deng, Yuan-Ming Li, Kun-Yu Lin, Wei-Shi Zheng

- **Overview:** Reference-space plans may look optimal but execute unstably or off-beat under whole-body control (WBC)—a reference-to-execution gap. Since fixed-window planning cannot adapt to variable rhythmic/phrase structures and reference quality misrepresents execution, ChoreoPlan combines beat-aligned variable-length discrete-continuous phrase planning with an Embodied Selector trained on offline WBC rollouts to preserve executability, rhythm, and semantics.

Modeling Multiple Normal Action Representations for Error Detection in Procedural Tasks
IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR 2025)

Wei-Jin Huang, Yuan-Ming Li, Zhi-Wei Xia, Yu-Ming Tang, Kun-Yu Lin, Jian-Fang Hu, Wei-Shi Zheng

- **Overview:** Procedural error detection must handle multiple valid next steps after the same history. Fixed temporal rules or a single static prototype ignore these futures and may select the wrong comparison target as execution changes. AMNAR predicts all valid next actions, reconstructs an adaptive normal representation for each, matches the current action to the best candidate, and thresholds their distance—avoiding incorrect prototypes under ambiguous futures.

EgoExo-Fitness: Towards Egocentric and Exocentric Full-Body Action Understanding
European Conference on Computer Vision (ECCV 2024)

Yuan-Ming Li, **Wei-Jin Huang (co-first author)**, An-Lan Wang, Ling-An Zeng, Jing-Ke Meng, Wei-Shi Zheng

- **Overview:** Ego/Exo understanding lacks synchronized data and fine-grained labels for what/when/how well. EgoExo-Fitness records 1,276 sequences from 40 participants with six synchronized cameras, annotated with two-level temporal boundaries, technical keypoint verification, natural-language comments, and quality scores. It benchmarks classification/localization, cross-view sequence verification and skill determination, and guidance-based execution verification.

TACD: Distilling Efficient Text-to-Motion Models via Terminal Amplification Control
arXiv preprint, 2026 · arXiv:2610.02867

Wei-Jin Huang, Yuan-Ming Li, Kun-Yu Lin, Wang Luo, Yinlin Zhu, Yue Yu, Shenghao Ye, Junbin Yuan, Fa-Ting Hong, Qing Zhang, Wei-Shi Zheng

- **Overview:** TACD accelerates text-to-motion generation by tying teacher query times to student step size, bounding excessive endpoint loss weights during few-step distillation. Training uses only text prompts and teacher supervision. On HumanML3D, the eight-step HY-Motion student reduces FID by 58% versus distillation without this bound. With a compact backbone and text encoder, it achieves a $7.7\times$ end-to-end speedup and reduces peak GPU memory from 17.63 to 2.62 GB. Retargeted motions execute on a Unitree G1 via whole-body control.